EFY Note
The source code of this project is available for free download at *source.efymag.com*

PARTS LIST

Semiconductors:	
Board1	- Arduino Uno
Miscellaneous:	
US1	- HC-SR04 ultrasonic sensor module
M1	- 5V servo motor MG90S
	- 9V PP3 battery
	- 2.1mm DC plug
	- Jumper wires
	- Dust-bin with cover

Fig. 4: Another view of author's installation

of the dust bin's lid, needs to be set through trial and error.

The project can be extended by using load cell and GPS sensor. A load cell can sense the weight of the bin. When the bin gets full and its weight exceeds certain limit, it will send alert signal to the nearest municipality and concerned garbage collector through the GPS. The exact location of the dust-bin can be tracked through the GPS latitude and longitude coordinates on Google Maps.

The circuit uses the software program loaded into the internal memory of Arduino Uno. The program *dustbin_EFY.ino* is simple and easy to understand. Comments are given at the end of each command line. The code starts by defining the pin numbers of ultrasonic sensor. The Trigger pin (input) and Echo pin (output) of the sensor are defined in the code.

Construction and testing

The circuit can be assembled on a general-purpose PCB. Before connecting the sensor and servo to Arduino, upload source code *dustbin_EFY.ino* to Arduino via Arduino IDE. To power up Arduino Uno board, a 9V PP3 battery with a 2.1mm DC plug can be used.

House the circuit in a suitable box and fix appropriately in the dust-bin. The installation in author's prototype during testing is shown in Fig. 3 and Fig. 4. EFY

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